

A thick, light gray curved line that starts from the top left, curves around the left side of the slide, and ends at the bottom left, separating the white background from the dark gray background.

χ^2

Chi-squared Test

Recall from lectures...

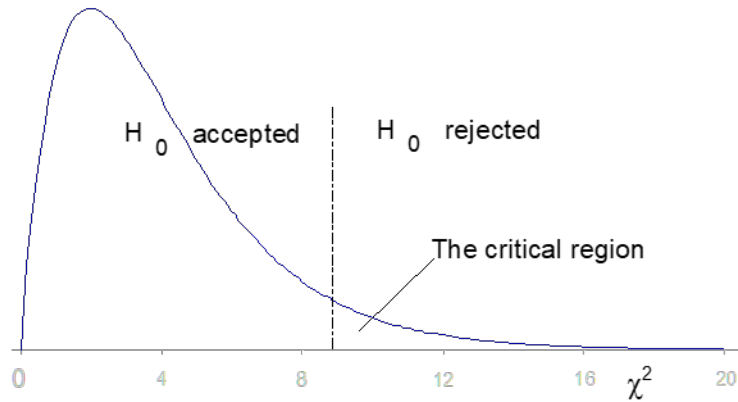
Chi-squared test:

- i. Goodness of fit test: to investigate the distribution of one categorical (nominal or ordinal) variable
- ii. Test of association: to see if there is any association between two categorical variables

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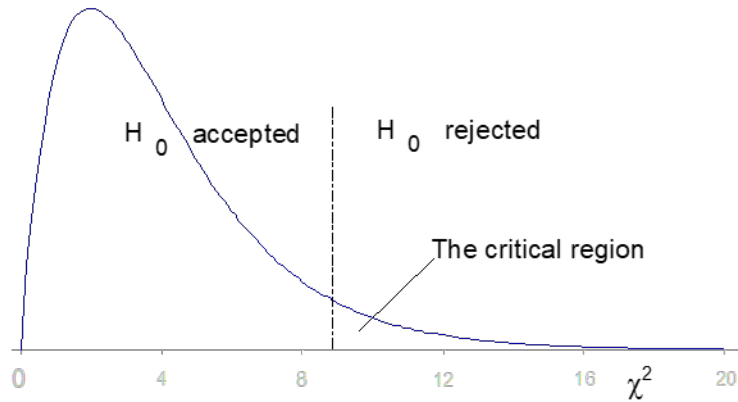
* the critical value depends on the degrees of freedom

* Test Statistic: $\chi^2 = \sum \frac{(O - E)^2}{E}$, O: observed, E: expected value

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Example: We throw a die 120 times, so we'd **expect** 20 of each number with random rolling. We

observe:

Number	1	2	3	4	5	6
Frequency	20	10	10	20	20	40

 Are the frequencies obtained close enough for random rolling?

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- 1. Null hypothesis: no difference between this die and a “fair” one
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observe: *Are the frequencies obtained close enough for random rolling?*

- 1. Null hypothesis: no difference between this die and a “fair” one
- 2. Alternative hypothesis: there is a significant difference between this die and a “fair” one
- 3. We need both the observed and the expected frequencies (remember $\chi^2 = \sum \frac{(O - E)^2}{E}$). Hence,

Number	1	2	3	4	5	6
Observed	20	10	10	20	20	40
Expected	20	20	20	20	20	20

⇒ $H_0: O = E \quad vs \quad H_1: O \neq E$

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 $\Rightarrow H_0: O = E \quad vs \quad H_1: O \neq E$
4. Calculate the test statistic of each column ($TS = \sum \frac{(O - E)^2}{E}$)
...and add up across the TS row: Test Statistic = 30

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Expected	20	20	20	20	20	20
TS	0	5	5	0	0	20

...and add up across the TS row: Test Statistic = 30

5. Critical Value: The row to look at is $n-1$, so 5 in this case, and again default to 5%. CV = 11.070 and $TS > CV$, so we reject the null and we conclude that this die is not fair

Recall from lectures...

- i. [Goodness of fit test Example](#)
- ii. [Test of association:](#)

Example: We ask students to evaluate the following sentence: *I am confident with numbers.*

Confident with Numbers	Gender
Disagree	F
Neither	M
Disagree	F
Neither	M
Neither	F
Agree	M
...	...

and the SPSS output gives us

Gender * I am confident with numbers Crosstabulation					
Count		I am confident with numbers			Total
		Disagree	Neither	Agree	
Gender	Male	8	25	31	64
	Female	23	29	43	95
Total		31	54	74	159

The null hypothesis is that there is NO correlation (in other words, they are independent of one another). If two variables are independent, we learn **NOTHING** about one from the other.

Recall from lectures...

1.

Observed	Disagree	Neither	Agree	Total
Male	8	25	31	64
Female	23	29	43	95
Total	31	54	74	159

$64/159 = 0.4$ and $95/159 = 0.6$. Hence, this 40-60 ratio should also be found in the three columns if we learn nothing from confidence with numbers by gender.

Recall from lectures...

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$64/159 = 0.4$ and $95/159 = 0.6$. Hence, this 40-60 ratio should also be found in the three columns if we learn nothing from *confidence with numbers* by gender.

2. We split 31, 54, and 74 in the above ratio:

Expected	Disagree	Neither	Agree	Total
Male	12.5	21.7	29.8	64
Female	18.5	32.3	44.2	95
Total	31	54	74	159

... $31 \times 0.4 = 12.5$ and $31 \times 0.6 = 18.5$, etc. This table says that all columns represent gender categories of 40/60 ratio.

Recall from lectures...

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3. Test Statistic: $TS = \sum \frac{(O-E)^2}{E}$

$TS_{M,D} = \frac{(8-12.5)^2}{12.5} = 1.61, TS_{F,D} = \frac{(23-18.5)^2}{18.5} = 1.08, \text{ etc.}$
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TS	Disagree	Neither	Agree	Total
Male	1.61	0.49	0.05	
Female	1.08	0.33	0.03	
Total				3.59

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TS	Disagree	Neither	Agree	Total
Male	1.61	0.49	0.05	
Female	1.08	0.33	0.03	
Total				3.59

Test Statistic = 3.59

4. Critical Value: In general, data with r rows and c columns will have $(r-1) \times (c-1)$ degrees of freedom. This only applies to situations with two variables.

* $r = 2$, $c = 3$, thus we have 2 degrees of freedom. CV = 5.991 at 5%

* $TS < CV$, we fail to reject the null, and we conclude that we cannot see any significant difference between the Observed and the Expected, namely Gender and Confidence with Numbers are not significantly associated.

Question 1

The two types of chi-squared test:

- i. are both for two variables
- ii. are the "goodness of fit" and the "test of association".
- iii. test for the individual significance of a variable
- iv. Perform an ANOVA test

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Question 2

A six-sided die is thrown a number of times, and the results are in the table below.

Face showing:	1	2	3	4	5	6
Frequency:	73	74	43	63	42	90

What frequency would we expect to see for each face were it a fair die?

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How many times do we throw the die?

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How many times do we throw the die? $73 + 74 + 43 + 63 + 42 + 90 = 385$

Then, a fair die shows up each face the same number of times:

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Bonus question: What is test statistic?

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Bonus question: What is test statistic?

$$TS = \sum \frac{(O-E)^2}{E} = 27.78$$

Question 3

A national vote has 28% of the voters for a proposal, and the remainder against. Locally, the 1,508 voters had 614 of them vote for it, and the remainder against.

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Local Vote	Positive	Negative
Observed	614	
Expected		

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Local Vote	Positive	Negative
Observed	614	894
Expected		

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Observed	614	894
Expected	422.24	

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TS + =

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Local Vote	Positive	Negative
Observed	614	894
Expected	422.24	1,085.76
TS	87.08	+ 33.87

= 121

$$TS = \sum \frac{(O - E)^2}{E}$$

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A test of association chi-squared test on data formed in 4 rows by 3 columns takes its critical value from which row of the table?

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	10%	5%	2.5%	1%	0.5%	0.1%
1	2.706	3.841	5.024	6.635	7.879	10.828
2	4.605	5.991	7.378	9.210	10.597	13.816
3	6.251	7.815	9.348	11.345	12.838	16.266
4	7.779	9.488	11.143	13.277	14.860	18.467
5	9.236	11.070	12.833	15.086	16.750	20.515
6	10.645	12.592	14.449	16.812	18.548	22.458
7	12.017	14.067	16.013	18.475	20.278	24.322
8	13.362	15.507	17.535	20.090	21.955	26.124
9	14.684	16.919	19.023	21.666	23.589	27.877
10	15.987	18.307	20.483	23.209	25.188	29.588
11	17.275	19.675	21.920	24.725	26.757	31.264
12	18.549	21.026	23.337	26.217	28.300	32.909
13	19.812	22.362	24.736	27.688	29.819	34.528
14	21.064	23.685	26.119	29.141	31.319	36.123
15	22.307	24.996	27.488	30.578	32.801	37.697
16	23.542	26.296	28.845	32.000	34.267	39.252
17	24.769	27.587	30.191	33.409	35.718	40.790
18	25.989	28.869	31.526	34.805	37.156	42.312
19	27.204	30.144	32.852	36.191	38.582	43.820
20	28.412	31.410	34.170	37.566	39.997	45.315
21	29.615	32.671	35.479	38.932	41.401	46.797
22	30.813	33.924	36.781	40.289	42.796	48.268
23	32.007	35.172	38.076	41.638	44.181	49.728
24	33.196	36.415	39.364	42.980	45.559	51.179
25	34.382	37.652	40.646	44.314	46.928	52.620
26	35.563	38.885	41.923	45.642	48.290	54.052
27	36.741	40.113	43.195	46.963	49.645	55.476
28	37.916	41.337	44.461	48.278	50.993	56.892
29	39.087	42.557	45.722	49.588	52.336	58.301
30	40.256	43.773	46.979	50.892	53.672	59.703
31	41.422	44.985	48.232	52.191	55.003	61.098
32	42.585	46.194	49.480	53.486	56.328	62.487
33	43.745	47.400	50.725	54.776	57.648	63.870
34	44.903	48.602	51.966	56.061	58.964	65.247
35	46.059	49.802	53.203	57.342	60.275	66.619
36	47.212	50.998	54.437	58.619	61.581	67.985
37	48.363	52.192	55.668	59.893	62.883	69.346
38	49.513	53.384	56.896	61.162	64.181	70.703
39	50.660	54.572	58.120	62.428	65.476	72.055
40	51.805	55.758	59.342	63.691	66.766	73.402

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A test of association chi-squared test on data formed in 4 rows by 3 columns takes its critical value from which row of the table?

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Remember: data with r rows and c columns will have $(r-1) \times (c-1)$ degrees of freedom. This only applies to situations with two variables.

Hence, if $r = 4$ and $c = 3$, then $df = (4-1) \times (3-1) = 6$

What is the CV at 5%?

	10%	5%	2.5%	1%	0.5%	0.1%
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2	4.605	5.991	7.378	9.210	10.597	13.816
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34	44.903	48.602	51.966	56.061	58.964	65.247
35	46.059	49.802	53.203	57.342	60.275	66.619
36	47.212	50.998	54.437	58.619	61.581	67.985
37	48.363	52.192	55.668	59.893	62.883	69.346
38	49.513	53.384	56.896	61.162	64.181	70.703
39	50.660	54.572	58.120	62.428	65.476	72.055
40	51.805	55.758	59.342	63.691	66.766	73.402

Question 4

A test of association chi-squared test on data formed in 4 rows by 3 columns takes its critical value from which row of the table?

Remember: data with r rows and c columns will have $(r-1) \times (c-1)$ degrees of freedom. This only applies to situations with two variables.

Hence, if $r = 4$ and $c = 3$, then $df = (4-1) \times (3-1) = 6$

What is the CV at 5%?

CV = 12.592

	10%	5%	2.5%	1%	0.5%	0.1%
1	2.706	3.841	5.024	6.635	7.879	10.828
2	4.605	5.991	7.378	9.210	10.597	13.816
3	6.251	7.815	9.348	11.345	12.838	16.266
4	7.779	9.488	11.143	13.277	14.860	18.467
5	9.236	11.070	12.833	15.086	16.750	20.515
6	10.645	12.592	14.449	16.812	18.548	22.458
7	12.017	14.067	16.013	18.475	20.278	24.322
8	13.362	15.507	17.535	20.090	21.955	26.124
9	14.684	16.919	19.023	21.666	23.589	27.877
10	15.987	18.307	20.483	23.209	25.188	29.588
11	17.275	19.675	21.920	24.725	26.757	31.264
12	18.549	21.026	23.337	26.217	28.300	32.909
13	19.812	22.362	24.736	27.688	29.819	34.528
14	21.064	23.685	26.119	29.141	31.319	36.123
15	22.307	24.996	27.488	30.578	32.801	37.697
16	23.542	26.296	28.845	32.000	34.267	39.252
17	24.769	27.587	30.191	33.409	35.718	40.790
18	25.989	28.869	31.526	34.805	37.156	42.312
19	27.204	30.144	32.852	36.191	38.582	43.820
20	28.412	31.410	34.170	37.566	39.997	45.315
21	29.615	32.671	35.479	38.932	41.401	46.797
22	30.813	33.924	36.781	40.289	42.796	48.268
23	32.007	35.172	38.076	41.638	44.181	49.728
24	33.196	36.415	39.364	42.980	45.559	51.179
25	34.382	37.652	40.646	44.314	46.928	52.620
26	35.563	38.885	41.923	45.642	48.290	54.052
27	36.741	40.113	43.195	46.963	49.645	55.476
28	37.916	41.337	44.461	48.278	50.993	56.892
29	39.087	42.557	45.722	49.588	52.336	58.301
30	40.256	43.773	46.979	50.892	53.672	59.703
31	41.422	44.985	48.232	52.191	55.003	61.098
32	42.585	46.194	49.480	53.486	56.328	62.487
33	43.745	47.400	50.725	54.776	57.648	63.870
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Question 5

A researcher is investigating an association between Age and Preferred Music. Their sample results are given below:

	Pop	Metal	Classical
Young	37	43	46
Middle-Aged	13	48	10
Elderly	26	21	38

If these two variables were independent, what would the researcher expect to have in the Metal and Young entry?

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Then, $112 \times 0.447 = 50.04$

Question 6

A researcher is investigating how drink preferences change with age. Their sample results are given in the table below:

	Tea	Coffee
Younger	39	22
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What is the test statistic to measure the association between Age and Drink Preference?

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Observed	Tea	Coffee	Total
Younger	39	22	
Older	19	17	
Total			

Expected	Tea	Coffee
Younger	?	?
Older	?	?

$$TS = \sum \frac{(O - E)^2}{E}$$

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1. Find the “nothing-to-learn” ratios:

Expected	Tea	Coffee
Younger	?	?
Older	?	?

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Observed	Tea	Coffee	Total
Younger	39	22	61
Older	19	17	36
Total	58	39	97

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- 3. Find the TS by age and drink, and then sum ($TS = \sum \frac{(O-E)^2}{E}$)

$TS_{y,T} = \frac{(39-36.48)^2}{36.48} = 0.174$, $TS_{o,T} = \frac{(19-21.52)^2}{21.52} = 0.295$, $TS_{y,C} = \frac{(22-24.53)^2}{24.53} = 0.261$, $TS_{o,C} = \frac{(17-14.47)^2}{14.47} = 0.442$ Then, overall, $TS = 1.17$

Question 7

Which of the following is a correct interpretation of the following SPSS output?

- 1. The Expected Counts show what would be the case if the rows were double the columns.
- 2. A one-tailed p-value of 0.041 indicates we should reject the null hypothesis and conclude that there is significant association between the two variables.
- 3. 0% of cells having expected count less than 5 means the test is invalid.
- 4. Because the test statistic is greater than the df, we should reject the null hypothesis.

VAR00001 * VAR00002 Crosstabulation

Count

		VAR00002			
		0	1	2	Total
VAR00001	0	14	9	12	35
	1	10	35	11	56
	2	15	33	16	64
	3	10	24	16	50
	4	5	19	5	29
Total		54	120	60	234

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	16.132 ^a	8	.041
Likelihood Ratio	16.298	8	.038
Linear-by-Linear Association	.508	1	.476
N of Valid Cases	234		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 6.69.

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Count

		VAR00002			
		0	1	2	Total
VAR00001	0	14	9	12	35
	1	10	35	11	56
	2	15	33	16	64
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	4	5	19	5	29
Total		54	120	60	234

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Sum up

Chi-square test:

- a. goodness-of-fit
- b. association
- c. spss output